

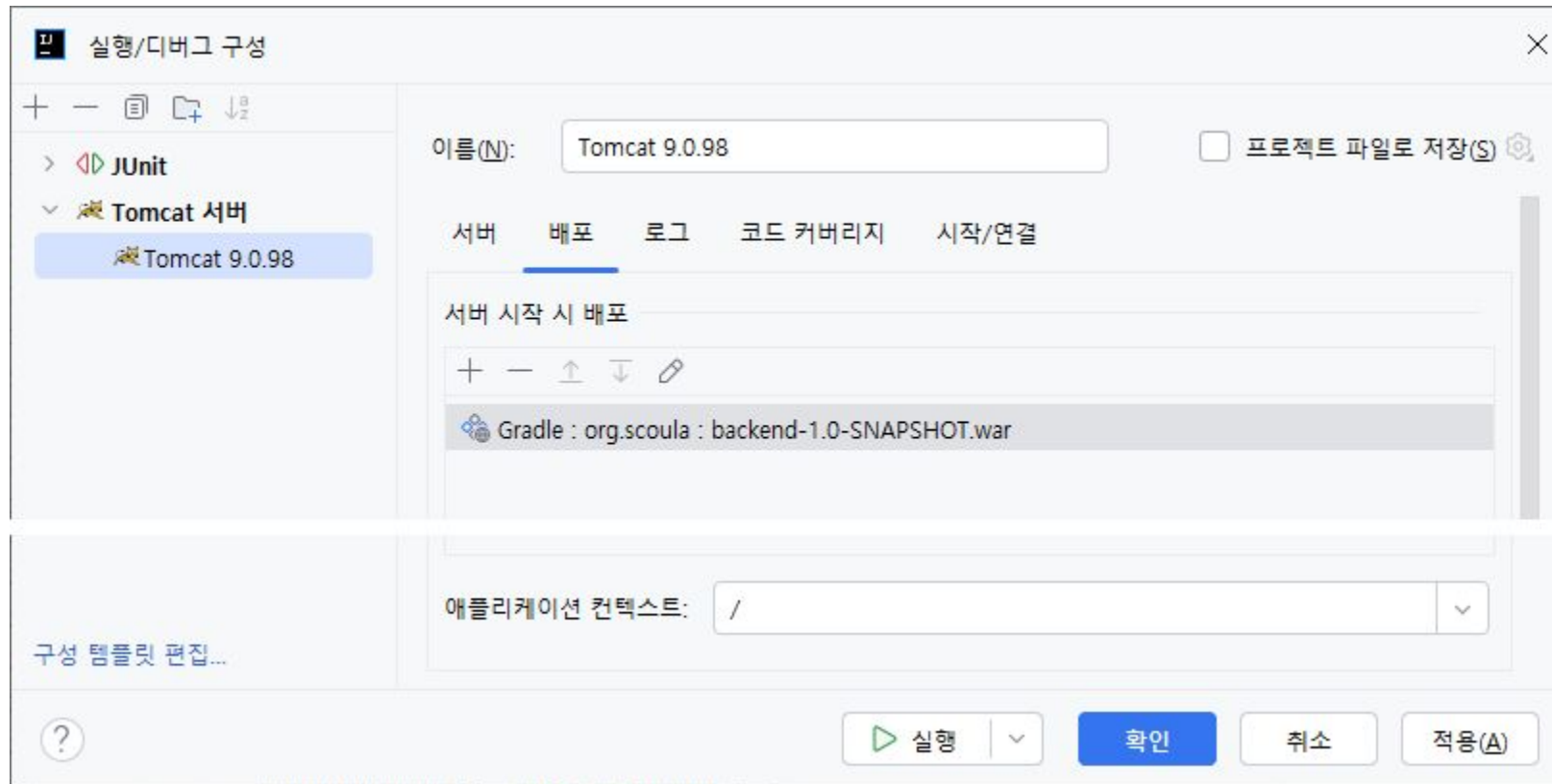
K-디지털 트레이닝

백엔드 준비

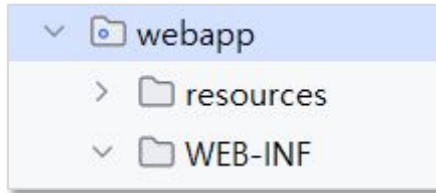
[KB] IT's Your Life

- **apiserver** 프로젝트를 **backend**로 복사
 - C:\KB_Fullstack\09_Spring+Vue\scoula\backend
 - **settings.gradle**에서 프로젝트 명 수정
`rootProject.name = "backend"`
 - **build.gradle**에 **swagger** 의존성 추가
`// swagger`
`implementation 'io.springfox:springfox-swagger2:2.9.2'`
`implementation 'io.springfox:springfox-swagger-ui:2.9.2'`
- Gradle Sync

- 실행 구성

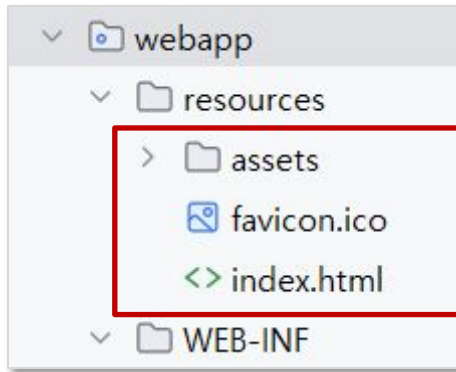


- webapp의 하위에 있는 모든 jsp 파일 삭제



- webapp/resources

- 프론트엔드에서 npm run build를 실행하면 배포되는 곳

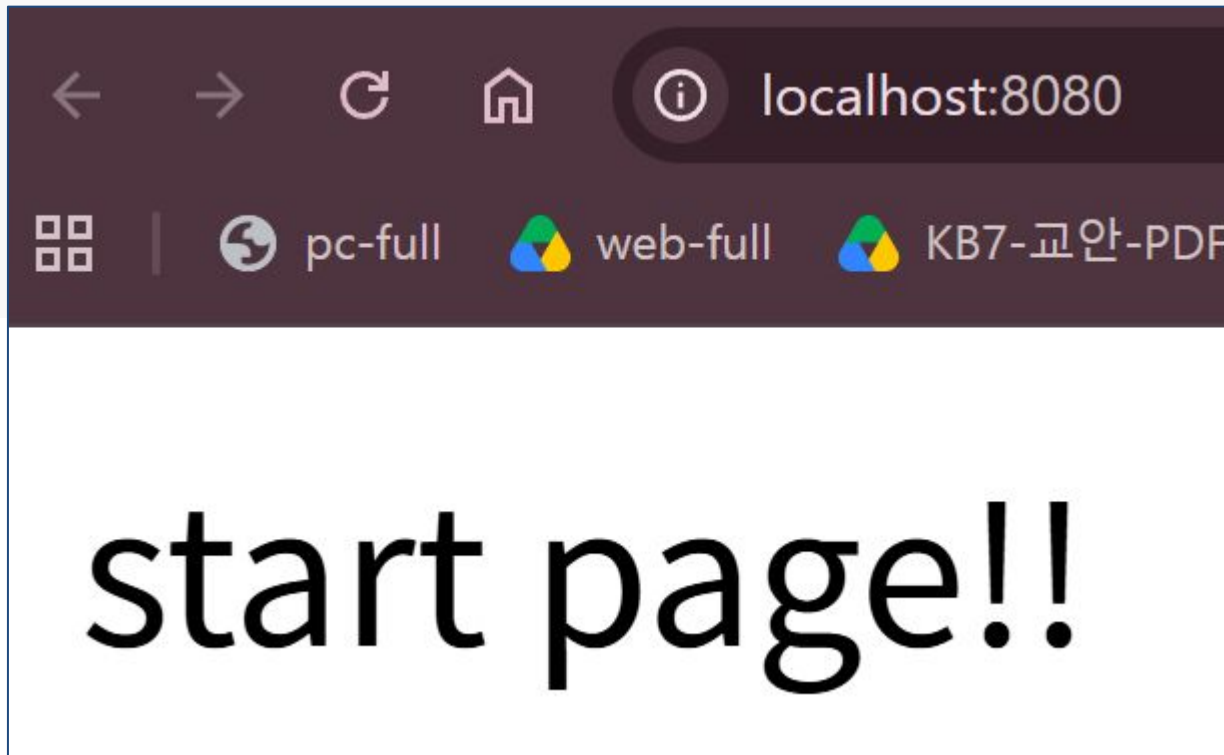
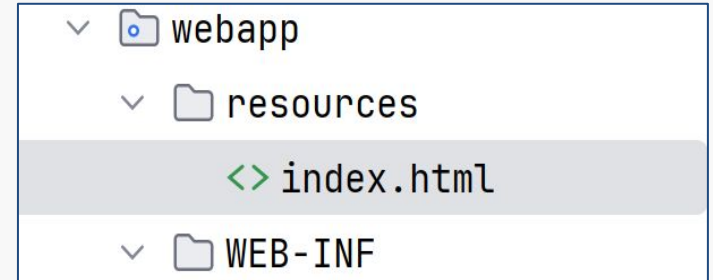


프론트엔드 build 결과물

- 기본 설정

- /를 요구했을 때 /resources/index.html을 리턴
- 프론트엔드에서 기존 정적 파일은 /assets/**로 기술되어 있음
 - 실제 배치는 /resources/assets/**
 - /assets/**를 요구했을 때 /resources/assets/** 경로로 해석하도록 설정
 - /assets/**는 스프링 처리에서 제외해야 함(resources에 대한 동일한 처리)
- 새로 고침했을 때 해당 url은 백엔드에 jsp파일 없음
 - 404에러 발생 → index.html로 포워딩
 - 일반 에러났을 때(404, 500등) CommonExceptionHandler에 지정한 jsp페이지를 부르도록 설정해두었음.
 - **jsp를 모두 삭제했으므로 index.html로 포워딩 설정**
 - 404등의 에러 발생시 예외처리하는 클래스 2개 설정해둠. → 적용 순서 지정
 - CommonExceptionHandler(일반 컨트롤러용, 특정한 jsp페이지 호출되도록 설정됨.)
 - ApiExceptionHandler(RestController용, json응답하도록 설정됨.)
 - **@Order(순번)**

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>Document</title>
</head>
<body>
start page!!
</body>
</html>
```



- **config.ServletConfig.java**

```
public class ServletConfig implements WebMvcConfigurer {  
    @Override  
    public void addViewControllers(ViewControllerRegistry registry) {  
        registry.addViewController("/")  
            .setViewName("forward:/resources/index.html");  
    }  
  
    @Override  
    public void addResourceHandlers(ResourceHandlerRegistry registry) {  
        registry.addResourceHandler("/resources/**")  
            .addResourceLocations("/resources/");  
  
        registry.addResourceHandler("/assets/**")  
            .addResourceLocations("/resources/assets/");  
    }  
    ...  
}
```

○ void configureViewResolvers(ViewResolverRegistry registry) 삭제

- **CommonExceptionHandler.java**

```
package org.scoula.exception;

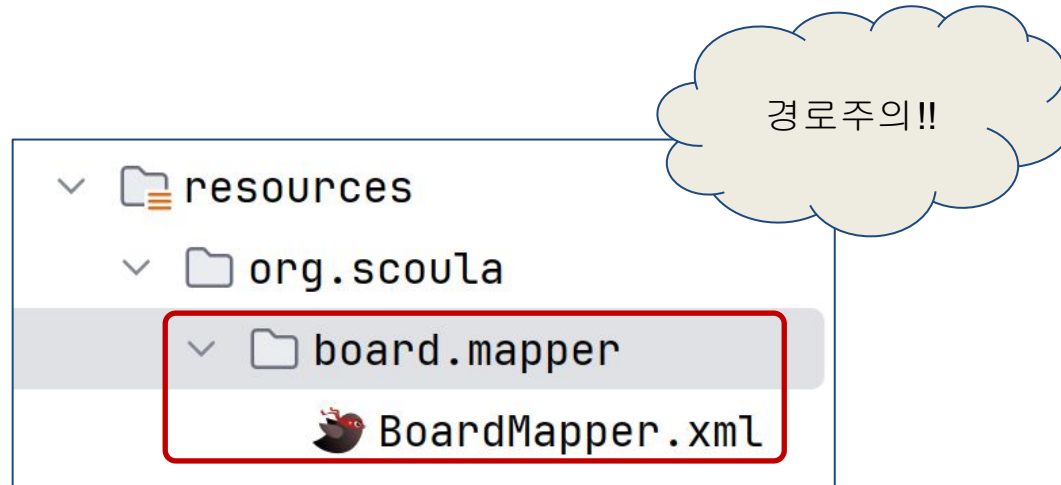
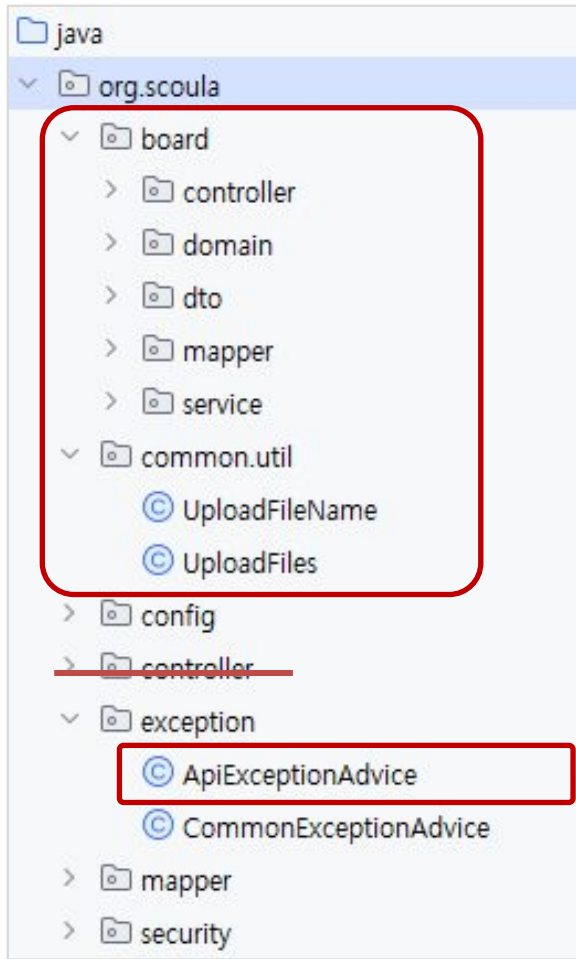
import lombok.extern.log4j.Log4j2;
import org.springframework.core.annotation.Order;
import org.springframework.web.bind.annotation.ControllerAdvice;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.servlet.NoHandlerFoundException;

@ControllerAdvice
@Log4j2
@Order(1)
public class CommonExceptionHandler {

    @ExceptionHandler(NoHandlerFoundException.class)
    public String handle404(NoHandlerFoundException ex) {
        return "/resources/index.html";
    }
}
```


- 게시판 (board) api 준비

- boardapi 프로젝트에서 **org.scoula.board** 패키지 복사
- exception.ApiExceptionAdvice.java, board.mapper.BoardMapper.xml** 복사



- **ApiExceptionAdvice.java**

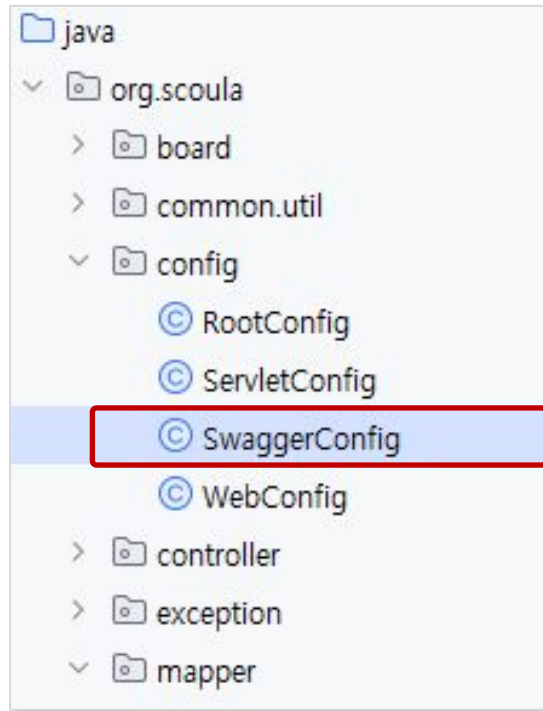
```
package org.scoula.exception;

import org.springframework.core.annotation.Order;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.RestControllerAdvice;

import java.util.NoSuchElementException;

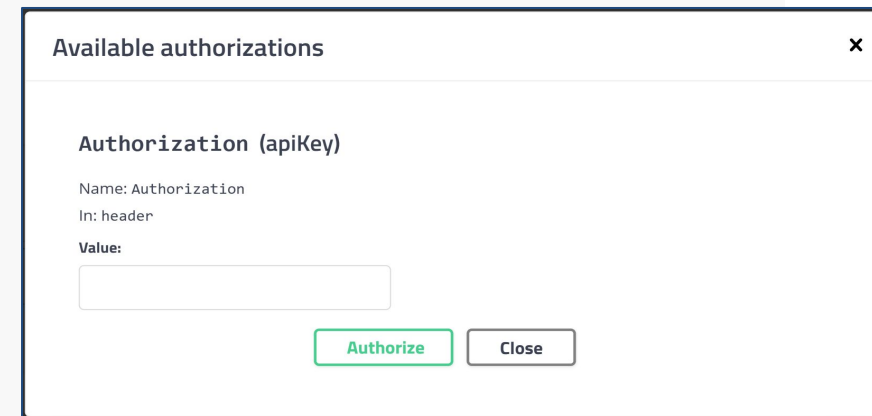
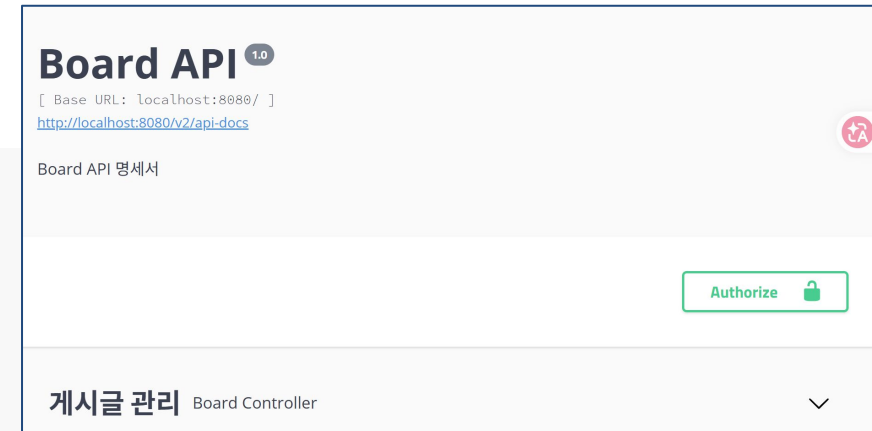
@RestControllerAdvice
@Order(2)
public class ApiExceptionAdvice {
    ...
}
```

- 게시판 (board) api 준비
 - SwaggerConfig.java 복사
 - JWT 인증 처리 추가



- **config.SwaggerConfig.java**

```
public class SwaggerConfig {  
    ...  
    @Bean  
    public Docket api() {  
        return new Docket(DocumentationType.SWAGGER_2)  
            // JWT 인증  
            .securityContexts(List.of(this.securityContext())) // SecurityContext 설정  
            .securitySchemes(List.of(this.apiKey())) // ApiKey 설정  
            .select()  
            .apis(RequestHandlerSelectors.withClassAnnotation(RestController.class))  
            .paths(PathSelectors.any())  
            .build()  
            .apiInfo(apiInfo());  
    }  
}
```



● config.SwaggerConfig.java

// JWT SecurityContext 구성

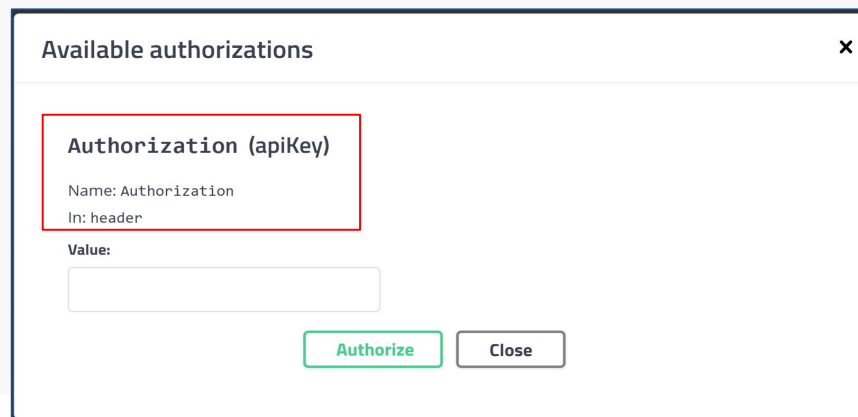
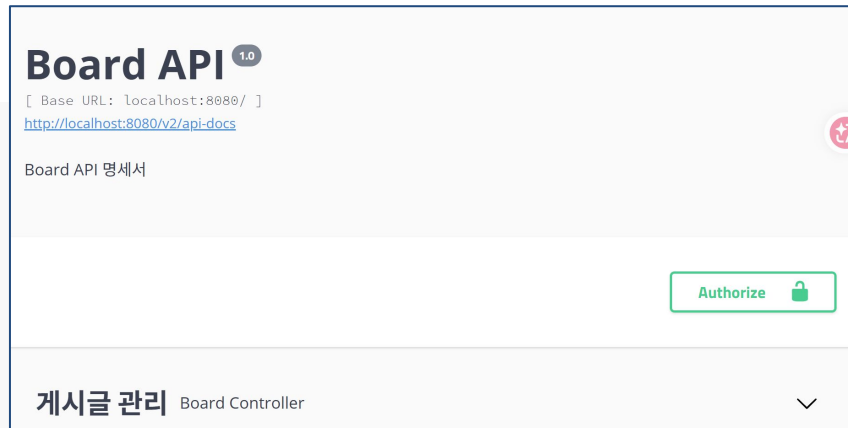
```
private SecurityContext securityContext() {
    return SecurityContext.builder()
        .securityReferences(defaultAuth())
        .build();
}
```

```
private List<SecurityReference> defaultAuth() {
```

```
    AuthorizationScope authorizationScope = new AuthorizationScope("global", "accessEverything");
    AuthorizationScope[] authorizationScopes = new AuthorizationScope[1];
    authorizationScopes[0] = authorizationScope;
    return List.of(new SecurityReference("Authorization", authorizationScopes));
}
```

// ApiKey 정의

```
private ApiKey apiKey() {
    return new ApiKey("Authorization", "Authorization", "header");
}
```



- **config.ServletConfig.java**

```
public class ServletConfig implements WebMvcConfigurer {  
    ...  
    @Override  
    public void addResourceHandlers(ResourceHandlerRegistry registry) {  
        ...  
        // Swagger UI 리소스를 위한 핸들러 설정  
        registry.addResourceHandler("/swagger-ui.html")  
            .addResourceLocations("classpath:/META-INF/resources/");  
  
        // Swagger WebJar 리소스 설정  
        registry.addResourceHandler("/webjars/**")  
            .addResourceLocations("classpath:/META-INF/resources/webjars/");  
  
        // Swagger 리소스 설정  
        registry.addResourceHandler("/swagger-resources/**")  
            .addResourceLocations("classpath:/META-INF/resources/");  
  
        registry.addResourceHandler("/v2/api-docs")  
            .addResourceLocations("classpath:/META-INF/resources/");  
    }  
}
```

- **config.WebConfig.java**

```
public class WebConfig extends AbstractAnnotationConfigDispatcherServletInitializer {  
    ...  
  
    @Override  
    protected Class<?>[] getServletConfigClasses() {  
        return new Class[] { ServletConfig.class, SwaggerConfig.class };  
    }  
}
```

- **resources::/mybatis-config.xml**

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE configuration
    PUBLIC "-//mybatis.org//DTD Config 3.0//EN"
    "http://mybatis.org/dtd/mybatis-3-config.dtd">

<configuration>
  <settings>
    <setting name="mapUnderscoreToCamelCase" value="true" />
  </settings>

  <typeAliases>
    <typeAlias alias="BoardVO" type="org.scoula.board.domain.BoardVO" />
    <typeAlias alias="AuthVO" type="org.scoula.security.account.domain.AuthVO" />
    <typeAlias alias="MemberVO" type="org.scoula.security.account.domain.MemberVO" />
  </typeAliases>

</configuration>
```


- **config.RootConfig.java**

```
@Configuration
@PropertySource({"classpath:/application.properties"})
@MapperScan(basePackages = {"org.scoula.board.mapper"})
@ComponentScan(basePackages = {"org.scoula.board.service"})
@Log4j2
@EnableTransactionManagement
public class RootConfig {
    ...
}
```

- **config.ServletConfig.java**

```
@EnableWebMvc
@ComponentScan(basePackages = {
    "org.scoula.controller",
    "org.scoula.exception",
    "org.scoula.board.controller"
})
public class ServletConfig implements WebMvcConfigurer {
    ...
}
```

- **security.config.SecurityConfig.java**

```
public class SecurityConfig extends WebSecurityConfigurerAdapter {  
    ...  
    // 접근 제한 무시 경로 설정 – resource  
    @Override  
    public void configure(WebSecurity web) throws Exception {  
        web.ignoring().antMatchers("/assets/**", "/*", "/api/member/**",  
            // Swagger 관련 url은 보안에서 제외  
            "/swagger-ui.html", "/webjars/**", "/swagger-resources/**", "/v2/api-docs"  
        );  
    }  
}
```

- **security.config.SecurityConfig.java**

```
@Override
```

```
public void configure(HttpSecurity http) throws Exception {
```

```
...
```

```
http
```

```
    .authorizeRequests()
```

```
    .antMatchers(HttpMethod.OPTIONS).permitAll()
```

```
    // 일단 모든 접근 허용
```

```
    .anyRequest().permitAll();
```

```
http.httpBasic().disable()           // 기본 HTTP 인증 비활성화
```

```
    .csrf().disable()               // CSRF 비활성화
```

```
    .formLogin().disable() // formLogin 비활성화 □ 관련 필터 해제
```

```
    .sessionManagement().sessionCreationPolicy(SessionCreationPolicy.STATELESS); // 세션 생성 모드 설정
```

```
}
```

```
...
```

```
}
```

<http://localhost:8080/api/board>

METHOD: GET SCHEME // HOST [":" PORT] [PATH ["?" QUERY]]

http://localhost:8080/api/board length: 31 byte(s)

Send

QUERY PARAMETERS

Response Cache Detected - Elapsed Time: 275ms

200

HEADERS ? pretty

X-Content-Type: nosniff
X-XSS-Protection: 1; mode=block
Cache-Control: no-cache, no-store, max-age=0, must-revalidate
Pragma: no-cache
Expires: 0
X-Frame-Options: DENY
Content-Type: application/json; charset=UTF-8
Transfer-Encoding: chunked
Date: Wed, 31 Jul 2024 08:59:22 GMT
Keep-Alive: timeout=20
Connection: keep-alive

COMPLETE REQUEST HEADERS

BODY ? pretty

```
[
  {
    no: 127,
    title: "페이징 테스트 110",
    content: "페이징 테스트 110",
    writer: "user0",
    regDate: 1722404772000,
    updateDate: 1722404772000,
    attaches: null,
    files: null
  },
  {
    no: 126,
```